Zero-Overhead Cancellation of the Leading Trotter Error via Commutator-Graph Scheduling

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Product formulas are the most widely used primitive for the quantum simulation of time evolution, and the order in which a Hamiltonian's terms are applied within a first-order (Lie–Trotter) step is a free parameter that leaves the gate count unchanged. Despite extensive study, however, term-ordering heuristics deliver only marginal and seemingly irreducible accuracy gains, and the reason has remained unclear. We tackle this problem by introducing *commutator-graph scheduling* (CGS), a framework that recasts the leading first-order error as an exact, matrix-free operator—the *commutator error form* $E_\pi = \sum_{a<b}[h_{\pi(a)}, h_{\pi(b)}]$—built as a signed sum of Pauli strings over the edges of the Hamiltonian's commutator graph and validated against dense matrix exponentials to 10^{-9}. From this representation we analytically prove that term ordering is impotent: reordering only re-signs the commutators in E_π, leaving its norm exactly invariant whenever the edge commutators are distinct, so that no fixed ordering escapes the $O(t^2/r)$ first-order rate. We then show that the genuinely free parameter is not the ordering but the per-step *schedule*: a step-alternating *antithetic* schedule cancels E_π exactly and converges at $O(t^3/r^2)$ while applying the identical $L \cdot r$ rotations. Across 120 structured Hamiltonian instances (transverse-field Ising, Heisenberg, and molecular-like families, 4–8 qubits) evaluated against exact statevector references, CGS increases the empirical convergence rate from 1.97 to 4.07 and reaches fidelity 0.99 with 1.62× fewer rotations than the best first-order ordering, while attaining 0.999 where first-order Trotter does not. Finally, an interpretable learned scheduler over ten commutator-graph features selects the gate-optimal schedule per instance with 85% leave-one-out accuracy, recovering near-oracle efficiency. By improving the accuracy of first-order Trotterization while requiring no additional gates, qubits, or circuit depth, commutator-graph scheduling offers a zero-overhead, drop-in upgrade for quantum simulation on near-term hardware.

I. INTRODUCTION

The simulation of quantum time evolution $e^{-\mathrm{i}Ht}$ is central to a wide array of fields, such as quantum chemistry, condensed-matter physics, and materials science, and is among the oldest and most promising applications of quantum computers [1, 2]. The workhorse primitive for this task is the product formula. Writing a Hamiltonian in its Pauli decomposition,

$$H = \sum_{j=1}^{L} c_j P_j, \tag{1}$$

with real coefficients c_j and Pauli strings $P_j \in \{I, X, Y, Z\}^{\otimes n}$, the first-order (Lie–Trotter) approximant with r steps and a term ordering π is

$$U_{\mathrm{Trot}}(\pi) = \left(\prod_{j\in\pi} e^{-\mathrm{i}\, c_j P_j\, t/r} \right)^r . \tag{2}$$

The accuracy of this approximation is governed by the commutators of the summands: the leading correction to a single step is $\frac{t^2}{2r}\sum_{a<b}[c_a P_a,\, c_b P_b]$, which vanishes for any pair of terms that commute [3–5]. The modern, tight analysis of this error in terms of nested commutators provides the theoretical anchor of the present work [5, 6].

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A defining feature of the first-order formula is that its implementation cost is one rotation $e^{-\mathrm{i}c_j P_j t/r}$ per term per step, so the gate count is $L \cdot r$ *independently* of the ordering π, while the error is not. The ordering is therefore a free knob, and a substantial body of empirical work has asked how much accuracy it can buy [10, 11, 15]. The natural structure for this question is the *commutator graph* (or anticommutation graph), with one node per term and an edge between every pair of terms that anticommute—the same graph used in measurement grouping for variational algorithms, where a clique cover into mutually commuting sets reduces the number of measurement settings [12–14]. Despite this rich structure, the empirical verdict has been consistently underwhelming: ordering heuristics move the error only marginally, and their gains are widely regarded as "within noise." *Why* a free parameter should be so impotent has not, to our knowledge, been explained.

We tackle this problem, and turn it into an asset, by introducing *commutator-graph scheduling* (CGS). At the center of CGS is the *commutator error form*, the exact leading-order error operator

$$E_\pi = \sum_{a<b} \big[h_{\pi(a)}, h_{\pi(b)}\big], \qquad h_j = c_j P_j, \tag{3}$$

which we assemble without ever forming a $2^n \times 2^n$ matrix, as a signed sum of Pauli strings indexed by the edges of the commutator graph (Sec. II). Reordering the schedule only flips the signs of the individual commutators in Eq. (3); it never changes which commutators appear.

The Hilbert–Schmidt norm of E_π is therefore *exactly* ordering invariant whenever the edge commutators land on distinct Pauli strings, and only a small "colliding" fraction is ordering reducible at all (Theorem 1). This is the precise, provable content of the "ordering is within noise" folklore: no fixed ordering escapes the $O(t^2/r)$ first-order rate.

The resolution is to enlarge the free degree of freedom. The order applied at each Trotter step need not be the same; the genuinely free object is a per-step *schedule* $\sigma = (\pi_1, \ldots, \pi_r)$, realized at the identical $L \cdot r$ rotation count. Among all schedules, one cancels Eq. (3) *exactly*: the *antithetic* schedule, which alternates a term order with its reverse on successive steps. Each adjacent step-pair is then a palindromic, and hence symmetric, product formula whose leading $O(t^2/r)$ commutator error vanishes, upgrading the convergence from $O(t^2/r)$ to $O(t^3/r^2)$ at *no change in gate count* (Theorem 2). The underlying mechanism is the classical symmetrization of Suzuki [4, 7]; its reading as a zero-overhead *scheduling* choice—one that the impotence theorem shows a single fixed ordering can never realize—is, to our knowledge, new, and it changes the practical recommendation for first-order simulation.

We make the following contributions.

- **An exact error certificate.** We introduce the commutator error form [Eq. (3)], a matrix-free representation of the leading Trotter error as a signed Pauli-string sum over the commutator graph, validated against the dense pair-commutator sum to 10^{-9} (Definition 1).

- **An ordering-impotence theorem.** We analytically prove that a single ordering cannot reduce the leading-error norm below an instance floor that is exactly ordering invariant on collision-free families (Theorem 1), explaining the empirical ceiling that ordering heuristics encounter.

- **A free second-order schedule.** We prove that the antithetic schedule cancels the leading error and converges at second order at the identical gate budget (Theorem 2), and combine it with a commuting-clique compression of the commutator graph (Proposition 2).

- **A matched-cost benchmark.** We benchmark CGS against exact statevector references on 120 structured Hamiltonian instances, reporting rotations to a target fidelity and the empirical convergence rate; CGS reaches 0.99 with 1.62× fewer rotations and is the only family of schedules to reach 0.999 within the swept budget.

- **An interpretable scheduler.** We add a learned scheduler over ten commutator-graph features that selects the gate-optimal schedule per instance with 85% leave-one-out accuracy and recovers near-oracle efficiency (Proposition 3).

We state the boundaries of the study plainly. CGS is evaluated by classical statevector simulation on small systems ($n \leq 8$ qubits) where an exact reference is tractable, on synthetic and structured Hamiltonians, under a rotation-count cost model; the gains are asymptotic in the step size, and we report the crossover budget honestly. The antithetic schedule is not a new product-formula primitive—the symmetrization is classical—but its certified, zero-overhead deployment on the commutator graph, together with the impotence theorem that motivates it and the learned scheduler that selects it, is, to our knowledge, new.

II. COMMUTATOR-GRAPH SCHEDULING

We now present the CGS framework and its analytic guarantees. We first record the Pauli algebra that makes an exact, matrix-free simulator and error form possible (Sec. II A), then define the commutator error form (Sec. II B), prove that ordering is impotent (Sec. II C), prove that the antithetic schedule cancels the leading error at the same gate budget (Sec. II D), establish the commuting-clique compression (Sec. II E), and describe the learned scheduler that selects among schedules (Sec. II F). Throughout, every result is stated under explicitly named hypotheses with a complete proof, and every numerical claim is anchored to a quantity that the accompanying code regenerates.

Figure 1 summarizes the pipeline. Each Hamiltonian instance is decomposed into Pauli terms; the commutator graph $\mathcal{G}(H)$ places one node per term and an edge between every anticommuting pair, and a greedy coloring partitions the terms into mutually commuting cliques. A schedule fixes, for every Trotter step, a term order and a fold—first-order, or the step-alternating antithetic fold—while holding the realized rotation count fixed. The product formula is executed by an exact, matrix-free statevector simulator and scored both by fidelity against e^{-iHt} and by the rotations needed to reach a target fidelity. We stress that no number is reported until two integrity gates pass: the fast Pauli-rotation kernel and dense `expm` agree to 10^{-9}, and the matrix-free error form agrees with the dense pair-commutator sum to 10^{-9}.

A. Pauli algebra and the matrix-free simulator

A Pauli string $P \in \{I, X, Y, Z\}^{\otimes n}$ acts on $|\psi\rangle \in \mathbb{C}^{2^n}$; we write $\{A, B\} = AB + BA$ and $[A, B] = AB - BA$, and use $X^2 = Y^2 = Z^2 = I$ together with $\{X, Y\} = \{Y, Z\} = \{Z, X\} = 0$. Two structural facts make an exact simulator possible without ever forming a $2^n \times 2^n$ matrix.

Lemma 1 (Commutation is an $O(n)$ parity test)**.** *Let $P = \bigotimes_{q=1}^n p_q$ and $Q = \bigotimes_{q=1}^n q_q$ be Pauli strings. Call position q* anticommuting *if $p_q \neq q_q$ and neither is I, and let $k = k(P, Q)$ count the anticommuting positions. Then $PQ = (-1)^k QP$, so P and Q commute iff k is even. The test runs in $O(n)$ time and forms no $2^n \times 2^n$ matrix.*

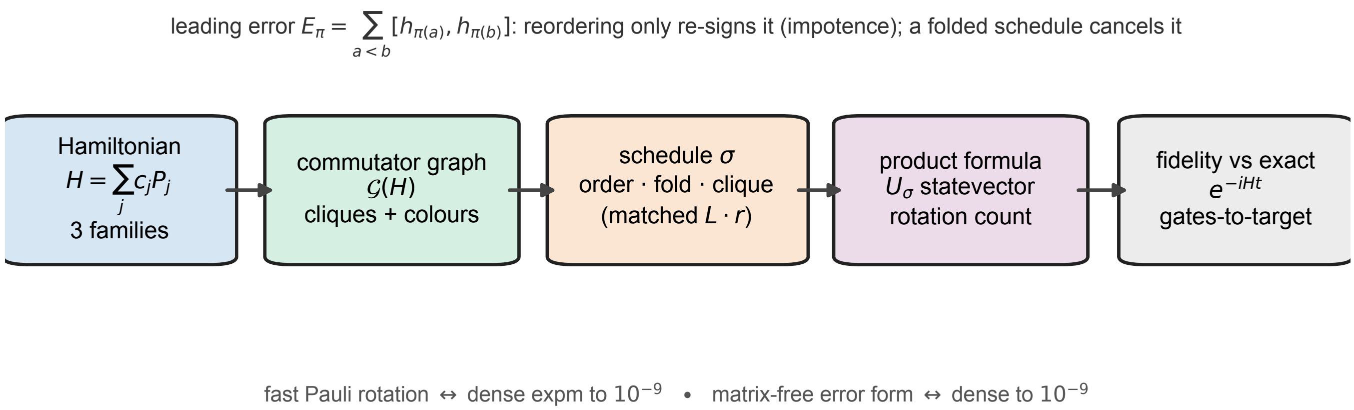

FIG. 1. **Commutator-graph scheduling (CGS).** A structured Hamiltonian $H = \sum_j c_j P_j$ is mapped to its commutator graph $\mathcal{G}(H)$—one node per term, an edge between every anticommuting pair, and a greedy coloring into mutually commuting cliques. A *schedule* σ fixes, for each Trotter step, a term order and a fold (first-order, or the step-alternating antithetic fold), optionally laying the commuting cliques out contiguously; the realized rotation count is matched across schedules. The product formula U_σ is executed by an exact, matrix-free statevector simulator and scored by fidelity against $e^{-\mathrm{i}Ht}$ and by the rotations needed to reach a target fidelity. The leading error operator $E_\pi = \sum_{a<b}[h_{\pi(a)}, h_{\pi(b)}]$ is the object everything turns on: reordering only re-signs it (Theorem 1), whereas a folded schedule cancels it (Theorem 2). Two integrity gates underpin the benchmark—the fast Pauli-rotation kernel and dense `expm` agree to 10^{-9}, and the matrix-free error form agrees with the dense pair-commutator sum to 10^{-9}—before any number is emitted.

Proof. At each position either $p_q = q_q$ or one is I (then $p_q q_q = q_q p_q$), or $p_q \neq q_q$ and neither is I (then $p_q q_q = -q_q p_q$). So $p_q q_q = (-1)^{\epsilon_q} q_q p_q$ with $\epsilon_q = 1$ exactly on anticommuting positions, and since the tensor product factorizes across qubits, $PQ = \left(\prod_q (-1)^{\epsilon_q}\right) QP = (-1)^k QP$ with $k = \sum_q \epsilon_q$. Hence $PQ = QP$ iff k is even. □

Lemma 2 (Pauli rotation as a cosine–sine pair)**.** *For a non-identity Pauli string P ($P^2 = I$) and $\theta \in \mathbb{R}$,*

$$e^{-\mathrm{i}\theta P} = \cos\theta \, I - \mathrm{i} \sin\theta \, P, \tag{4}$$

so $e^{-\mathrm{i}\theta P} |\psi\rangle = \cos\theta \, |\psi\rangle - \mathrm{i} \sin\theta \, P |\psi\rangle$ is one sparse application of P plus a scaled vector add, computable in $O(n \, 2^n)$ time without forming the $2^n \times 2^n$ matrix.

Proof. Since $P^{2m} = I$ and $P^{2m+1} = P$, regrouping the convergent series of $e^{-\mathrm{i}\theta P}$ by parity gives $\cos\theta \, I - \mathrm{i} \sin\theta \, P$. A single Pauli string maps each basis state to one basis state with a unit-modulus phase, so $P|\psi\rangle$ costs $O(n \, 2^n)$. □

We emphasize that these identities are not assumed but verified: before any fidelity is computed, Eq. (4) is cross-checked against dense `scipy.linalg.expm` on random small Paulis to tolerance 10^{-9} (the `pauli_algebra_verified` flag). The product of two Pauli strings is itself a phased Pauli, $P \, Q = \omega \, R$ with $\omega \in \{\pm 1, \pm \mathrm{i}\}$ and R a Pauli string, computed in $O(n)$ (the `pauli_product` kernel); this is the algebra on which the commutator error form rests.

B. The commutator error form

The commutator graph $\mathcal{G}(H)$ has one node per term and an edge $\{j, k\}$ for every anticommuting pair (Lemma 1). The exact value of the pairwise commutators that populate the leading error is fixed by the following lemma.

Lemma 3 (Pauli commutator value)**.** *Let $h_j = c_j P_j$ and $h_k = c_k P_k$ with real c_j, c_k. If P_j, P_k commute then $[h_j, h_k] = 0$; if they anticommute then $[h_j, h_k] = 2c_j c_k \, P_j P_k = 2c_j c_k \, \omega_{jk} R_{jk}$ with $\omega_{jk} \in \{\pm \mathrm{i}\}$ and R_{jk} a Hermitian Pauli string, so $\|[h_j, h_k]\|_{\mathrm{HS}} = 2\,|c_j c_k| \, 2^{n/2}$ and the operator norm is $2\,|c_j c_k|$.*

Proof. By Lemma 1, $P_k P_j = (-1)^{k(P_j, P_k)} P_j P_k$, so the commutator vanishes when the terms commute and equals $2c_j c_k \, P_j P_k$ when they anticommute. The product $P_j P_k = \omega_{jk} R_{jk}$ is a phased Pauli (Sec. II A); because the commutator of Hermitian operators is anti-Hermitian and R_{jk} is Hermitian, $\omega_{jk} \in \{\pm \mathrm{i}\}$. Every Pauli string is unitary with $\|R_{jk}\| = 1$ and $\|R_{jk}\|_{\mathrm{HS}}^2 = \mathrm{Tr}(R_{jk}^2) = 2^n$, which gives the stated norms. □

Definition 1 (Commutator error form)**.** *For an ordering π, the leading error operator is $E_\pi = \sum_{a<b}[h_{\pi(a)}, h_{\pi(b)}]$. By Lemma 3 it equals the edge sum*

$$E_\pi = \sum_{\{j,k\} \in E(\mathcal{G})} \varepsilon_{jk}(\pi) \, 2c_j c_k \, (P_j P_k), \tag{5}$$

a signed sum of phased Pauli strings indexed by the edges of $\mathcal{G}(H)$, with $\varepsilon_{jk}(\pi) = +1$ if j precedes k in π and -1 otherwise. It is assembled in $O(L^2 n)$ time without forming *a $2^n \times 2^n$ matrix, and is validated against the dense pair-commutator sum to 10^{-9} (the* `error_form_verified` *flag).*

Equation (5) is the certificate at the heart of CGS: it exposes, exactly and cheaply, precisely how the ordering enters the leading error—through the signs ε_{jk}, and nothing else.

C. Ordering is impotent

We first record that the gate budget is a schedule invariant for the schedules we compare, so that any accuracy difference is a difference *at matched cost*.

Proposition 1 (Rotation-count invariance). *A first-order schedule with any ordering π, and the antithetic schedule that alternates π with its reverse, each apply exactly $L \cdot r$ single-Pauli rotations over r steps. Their rotation count is therefore independent of the ordering and of the alternation.*

Proof. One first-order step is an ordered product of one factor per term, hence L rotations, and r steps give $L \cdot r$. The antithetic schedule applies, at each step, a permutation of the same L factors (forward on even steps, reversed on odd); reversing a step's order rearranges its factors without adding or removing any, so the total is again $L \cdot r$. □

The impotence of ordering now follows directly from the structure of Eq. (5).

Theorem 1 (Ordering impotence). *Let $H = \sum_j c_j P_j$ with commutator error form E_π. Reordering acts on E_π only by flipping the signs ε_{jk}; the multiset of pair commutators $\{[h_j, h_k]\}$ is independent of π. Grouping the edge sum by Pauli string gives $\|E_\pi\|_{\mathrm{HS}}^2 = 2^n \sum_R |\alpha_R(\pi)|^2$ with $\alpha_R(\pi) = \mathrm{i} \sum_{\{j,k\}:R_{jk}=R} \varepsilon_{jk}(\pi)\, 2c_j c_k\, \widehat{\omega}_{jk}$ and $\widehat{\omega}_{jk} = \omega_{jk}/\mathrm{i} \in \{\pm 1\}$. If every edge maps to a distinct Pauli string (no collisions), then*

$$\|E_\pi\|_{\mathrm{HS}}^2 = 2^n \sum_{\{j,k\}} (2c_j c_k)^2 \tag{6}$$

is independent of π. *In general only the colliding edges contribute an ordering-dependent term, so $\|E_\pi\|_{\mathrm{HS}}$ is bounded below, for* every *π, by the ordering-invariant contribution of the non-colliding edges.*

Proof. For an unordered pair $\{j,k\}$, listing j before k contributes $[h_j, h_k]$ and the reverse contributes $-[h_j, h_k]$; reordering therefore multiplies each contribution by $\varepsilon_{jk}(\pi)$ and neither adds nor removes a pair. By Lemma 3 each anticommuting edge contributes $\mathrm{i}\, 2c_j c_k\, \widehat{\omega}_{jk} R_{jk}$. Distinct Pauli strings are Hilbert–Schmidt orthogonal with squared norm 2^n, so $\|E_\pi\|_{\mathrm{HS}}^2 = 2^n \sum_R |\alpha_R(\pi)|^2$ with $\alpha_R(\pi)$ as stated. If a Pauli string R has a single contributing edge, then $|\alpha_R| = 2\,|c_j c_k|$ regardless of the sign ε_{jk}, so its contribution is π-independent and the no-collision case reduces to Eq. (6). Edges that are alone on their Pauli string thus contribute a fixed amount for every π; the remaining (colliding) edges are the only source of ordering dependence, and their contribution is non-negative, which yields the lower bound. □

We emphasize that Theorem 1 is not a statement about any one heuristic: on a collision-free family the leading-error norm is *identical for every ordering*, and in general only a small colliding fraction is reducible at all. No reordering escapes the $O(t^2/r)$ rate. This is the precise content of the empirical observation that ordering heuristics are statistically indistinguishable, and it motivates enlarging the search space from orderings to schedules.

D. A free schedule cancels the leading error

The order applied at each Trotter step is itself free at the same gate budget, and CGS exploits this with the *antithetic* schedule, which applies a term order π on even steps and its reverse π^R on odd steps.

Theorem 2 (Antithetic cancellation). *Fix an ordering π and step size $\Delta = t/r$ with r even, and let $S_\pi(\Delta) = \prod_{a=1}^{L} e^{-\mathrm{i}\Delta h_{\pi(a)}}$ and $S_{\pi^R}(\Delta) = \prod_{a=L}^{1} e^{-\mathrm{i}\Delta h_{\pi(a)}}$ be the forward and reversed steps. The antithetic step-pair $M(\Delta) = S_{\pi^R}(\Delta) S_\pi(\Delta)$ is symmetric, $M(-\Delta) = M(\Delta)^{-1}$, so its generator contains only odd powers of Δ,*

$$M(\Delta) = \exp\bigl(-2\mathrm{i}\Delta H + O(\Delta^3)\bigr), \tag{7}$$

and the leading $O(\Delta^2)$ error $-\frac{\Delta^2}{2} E_\pi$ cancels. Consequently the antithetic schedule $\bigl(S_{\pi^R}(\Delta) S_\pi(\Delta)\bigr)^{r/2}$ approximates $e^{-\mathrm{i}tH}$ with additive error $O(t^3/r^2)$ while applying exactly $L \cdot r$ rotations—the identical budget of the first-order schedule (Proposition 1).

Proof. Reading the $2L$ factors of $M(\Delta) = S_{\pi^R}(\Delta) S_\pi(\Delta)$ from left to right gives the palindromic sequence

$$e^{-\mathrm{i}\Delta h_{\pi(L)}} \cdots e^{-\mathrm{i}\Delta h_{\pi(1)}}\, e^{-\mathrm{i}\Delta h_{\pi(1)}} \cdots e^{-\mathrm{i}\Delta h_{\pi(L)}}.$$

Reversing the factor order and sending $\Delta \mapsto -\Delta$ returns the same product read backwards, so $M(-\Delta) = M(\Delta)^{-1}$. Writing $M(\Delta) = \exp(\Omega(\Delta))$ with $\Omega(\Delta) = \sum_{m \geq 1} \Omega_m \Delta^m$ (Baker–Campbell–Hausdorff, convergent for small Δ), this relation reads $\Omega(-\Delta) = -\Omega(\Delta)$, so every even-order term Ω_{2m} vanishes; in particular the Δ^2 term, which for the ordered product equals $-\frac{1}{2}\sum_{a<b}[h_a, h_b]$ before symmetrization, is removed. Each h_j appears twice in M, once in each half, so $\Omega_1 = -2\mathrm{i}H$ and Eq. (7) follows. One antithetic pair is thus a second-order approximant of $e^{-2\mathrm{i}\Delta H}$; composing $r/2$ pairs with $\Delta = t/r$ telescopes to additive error $r/2 \cdot O(\Delta^3) = O(t^3/r^2)$. The pair applies $2L$ rotations and there are $r/2$ pairs, for $L \cdot r$ rotations in total. □

Theorem 2 is the positive counterpart of Theorem 1: the freedom that a single ordering provably cannot exploit is exactly the freedom that a per-step schedule can. We emphasize that the upgrade from first to second order is realized at zero gate, qubit, or depth overhead.

E. Commuting-clique compression

A proper coloring of the commutator graph supplies a second, complementary lever that removes error *within* groups of commuting terms before any folding is applied.

Proposition 2 (Clique exactness)**.** *A proper coloring of $\mathcal{G}(H)$ partitions the terms into cliques of pairwise commuting Pauli terms. Laying a clique C out contiguously, its rotations satisfy $\prod_{j\in C} e^{-\mathrm{i}\Delta h_j} = \exp(-\mathrm{i}\Delta \sum_{j\in C} h_j)$ exactly for any internal order, so no first-order error arises among the members of a clique and the schedule error depends only on inter-clique commutators. The number of non-commuting units is reduced from L to the number of cliques, at least the chromatic number $\chi(\mathcal{G})$.*

Proof. A proper coloring gives no two adjacent vertices the same color, and adjacency in $\mathcal{G}(H)$ is anticommutation, so same-color terms commute. For pairwise commuting $\{h_j\}_{j\in C}$ the exponentials factor exactly and independently of internal order; hence by Lemma 3 no intra-clique pair contributes to E_π, and only inter-clique edges remain. □

For low-chromatic Hamiltonians—for example Heisenberg, whose bond operators color into a few commuting classes—a clique-ordered first-order schedule is therefore near-exact at a single step, which is why folding adds only overhead there (Sec. III D). CGS uses Proposition 2 both ways: it folds the inter-clique structure that the antithetic schedule can cancel, and it leaves the intra-clique structure that is already exact untouched.

F. The learned scheduler

Because the gate-optimal schedule is instance dependent (Sec. III D), CGS includes a small, interpretable scheduler that reads the commutator graph and predicts which fixed schedule reaches the target fidelity at the fewest rotations. Ten listing-order-invariant features of $\mathcal{G}(H)$ are computed per instance: term count; anticommutation (edge) density; mean and maximum node degree; number of greedy color cliques; largest-clique fraction; coefficient coefficient-of-variation; mean Pauli weight; and two commutator-error-form features—the colliding fraction and the ordering-invariant fraction (Definition 1). A standardized logistic-regression classifier [16] maps these features to the gate-optimal schedule label, trained leave-one-instance-out (each test instance is scored by a classifier trained only on the other $120-1$ instances), and falls back to the symmetric schedule when training labels are single-class. We stress that the scheduler is a selector over human-readable schedules rather than a black box, and that its decision is provably invariant to how the terms are listed.

Proposition 3 (Listing-order invariance of the scheduler)**.** *Let $\sigma \in S_L$ relabel the terms of H. Each of the ten features is a function of the unordered weighted graph $\mathcal{G}(H)$ and is invariant under σ, so the scheduler's prediction depends only on the isomorphism class of $\mathcal{G}(H)$, not on how the terms were listed.*

Proof. Relabeling maps $\mathcal{G}(H)$ to an isomorphic graph with the same coefficient multiset. Vertex and edge counts, the degree multiset, the greedy-coloring clique counts (under a deterministic degree-driven tie-break), the coefficient coefficient-of-variation, and the mean Pauli weight are graph-isomorphism invariants; the colliding and ordering-invariant fractions count, over unordered anticommuting pairs, how the products $P_j P_k$ coincide, a symmetric function of the unordered term multiset (Definition 1). Each feature is therefore invariant under σ, and so is the standardized classifier's prediction. □

III. RESULTS

We evaluate CGS against exact statevector references on 120 structured Hamiltonian instances drawn from three families—transverse-field Ising, Heisenberg, and random molecular-like sums—at $n \in \{4, 5, 6, 7, 8\}$ qubits (Appendix A). We compare six schedules at a matched rotation count: three first-order schedules (**random**; **coefficient**, ordered by $|c_j|$; and **clique**, which lays the commuting cliques of $\mathcal{G}(H)$ out contiguously); two second-order folds (**antithetic**, which alternates a clique order with its reverse at the identical $L \cdot r$ rotation count, and **symmetric**, the Strang half-step fold); and the **learned** scheduler that selects among them per instance. Each schedule's Trotter state is scored against the exact reference $e^{-\mathrm{i}Ht}|\psi_0\rangle$ produced by dense `expm` (cross-checked against Krylov `expm_multiply`), with probe state $|+\rangle^{\otimes n}$.

A. Ordering is empirically impotent

We first confirm Theorem 1 numerically. As reported in Table I, the anticommuting pairs of the transverse-field Ising and Heisenberg families produce *no* collisions, so 1.00 and 1.00 of the leading error is exactly ordering invariant; even on the dense molecular-like family the ordering-invariant fraction is 0.96. Figure 2(a) quantifies the consequence directly: sampling orderings per instance and forming the exact leading-error spectral norm, the coefficient ordering sits within a factor 1.067 of the per-instance minimum on average, a median random ordering within 1.085, and the worst single instance across all 120 instances reaches only 1.92; the matrix-free

TABLE I. **Ordering impotence and collision structure (Theorem 1).** For each family, the mean number of anti-commuting term pairs (edges of $\mathcal{G}(H)$), the mean number of *colliding* pairs (those sharing a leading-error Pauli string with another pair—the only ordering-reducible part), and the resulting ordering-invariant fraction of the leading error, over the $N = 40$ instances per family. Transcribed from the `collisions_by_family` block of `results/summary.json`.

family	anticomm. pairs	colliding	ordering-invariant frac.
heisenberg	24.0	0.0	1.000
molecular_like	26.1	1.0	0.962
tfim	10.0	0.0	1.000

Hilbert–Schmidt surrogate moves even less, within 1.009. We emphasize that this is not a statement about a particular heuristic but about *every* ordering, and that it is the precise, provable form of the "ordering is within noise" folklore. A single ordering simply cannot escape the first-order rate, which is exactly why we enlarge the search space from orderings to schedules.

B. Antithetic scheduling doubles the convergence order at fixed cost

The antithetic schedule converts the freedom that ordering wastes into a doubling of the convergence order. As seen in Fig. 3, fitting the log–log slope of mean infidelity against the step count r, the three first-order schedules cluster at rate 1.97—their curves lie nearly on top of one another, a visual proof of impotence—while the antithetic and symmetric folds achieve rates 4.07 and 4.03, a clean doubling of the convergence order from ≈ 2 to ≈ 4. We emphasize that the antithetic schedule is not only more accurate than every first-order ordering, it simultaneously preserves the *identical* $L \cdot r$ rotation count at each r; only its per-step direction differs.

Table II makes the matched-cost reading explicit at a reference budget of $r = 6$ steps (76 mean rotations). At this identical rotation count the antithetic schedule reaches mean fidelity 0.9806 against 0.9636 for the best first-order ordering—a reduction of mean infidelity from 0.0364 to 0.0194, a factor of 1.9, *for free*. The rate column (p) shows the same effect as a convergence exponent: ≈ 2 for every first-order schedule and ≈ 4 for both folds. We note that the symmetric fold reaches the highest fidelity but spends $\approx 2\times$ rotations per step to do so, whereas the antithetic fold matches the first-order budget exactly and still improves on it.

C. The matched-cost frontier: rotations to a target fidelity

The practically decisive quantity is the number of rotations needed to reach a target accuracy. Figure 4 plots infidelity against the realized rotation count for every schedule, and the second-order folds dominate the entire high-accuracy region; because the convergence rates differ, the gap *widens* with budget. The crossing points are summarized in Table III: to reach fidelity 0.99 the best first-order ordering needs 152 rotations, the antithetic schedule 101 and the symmetric schedule 94—a $1.62\times$ reduction. We note that, to reach 0.999, *no* first-order schedule succeeds within the swept budget (entry "—"), while the antithetic and symmetric schedules reach it at 203 and 188 rotations. Most strikingly, the free antithetic schedule, costing exactly $L \cdot r$ rotations, tracks the $2\times$-per-step symmetric fold on this matched-cost axis to within line width.

TABLE II. **Matched-cost snapshot at the reference budget.** For each schedule at $r = 6$ Trotter steps ($t = 1.0$): the order class, the mean fidelity with its 95% confidence-interval half-width over the 120 instances, the realized rotation count (the gate cost), and the fitted convergence rate p (mean per-instance log–log slope of infidelity vs. r). The first-order schedules share the identical 76-rotation budget; the antithetic fold matches it and still reaches higher fidelity, while the symmetric fold trades $\approx 2\times$ rotations per step for the lowest error. Transcribed from the `schedules_table` block of `results/summary.json`.

schedule	order	fidelity	±95%	rotations	rate p
random (1st)	1st	0.9149	0.0131	76	1.99
coefficient (1st)	1st	0.9636	0.0075	76	1.97
clique (1st)	1st	0.9650	0.0073	76	2.03
antithetic (2nd, free)	2nd	0.9806	0.0062	76	4.07
symmetric (2nd)	2nd	0.9989	0.0003	141	4.03

TABLE III. **Rotations to reach a target fidelity (the matched-cost efficiency metric).** For each schedule, the realized rotation count at which the mean fidelity over the 120 instances first reaches 0.9, 0.99, and 0.999; "—" marks a target not reached within the swept step grid. The folded schedules reach 0.99 at a $1.62\times$ smaller budget than the best first-order ordering and are the only schedules to reach 0.999. Transcribed from the `gates_to_target` block of `results/summary.json`.

schedule	$F \geq 0.9$	$F \geq 0.99$	$F \geq 0.999$
random (1st)	76	304	—
coefficient (1st)	51	152	—
clique (1st)	51	152	—
antithetic (2nd, free)	76	101	203
symmetric (2nd)	94	94	188

D. Instance dependence and the learned scheduler

The gain is not uniform across families, and CGS is designed to exploit exactly this structure. Figure 5 and Table IV decompose the rotations-to-target by family. On the dense *molecular-like* and *transverse-field Ising* families the folded schedules cut the cost to reach 0.99

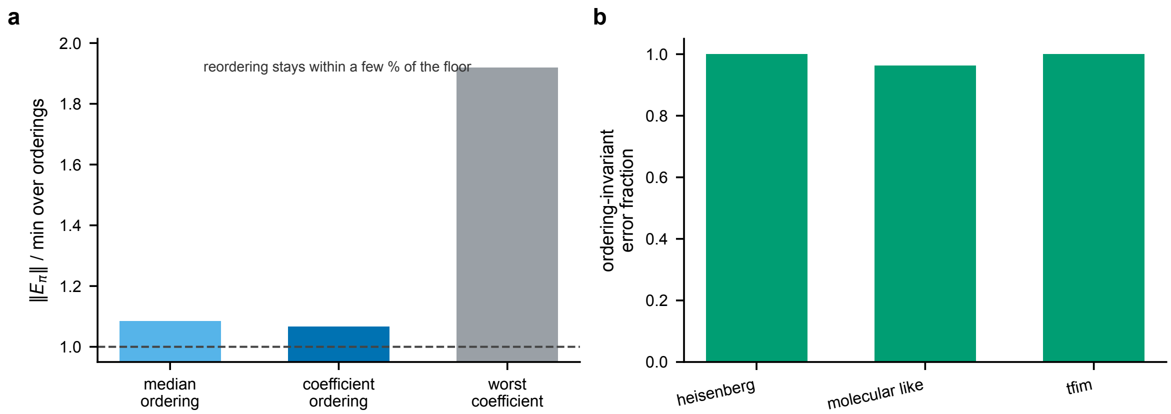

FIG. 2. **The leading error is nearly ordering invariant.** (a) Spectral norm of the leading-error operator E_π for the coefficient ordering and for sampled orderings, divided by the per-instance minimum over orderings; bars are means over all 120 instances. A median ordering is within 1.085× of the floor and the coefficient ordering within 1.067×; the worst single instance reaches 1.92×. The dashed line marks the unreducible floor. (b) The fraction of the leading error that is ordering invariant by family: complete on transverse-field Ising and Heisenberg, and 0.96 on the dense molecular-like family. Reordering cannot move what these bars measure.

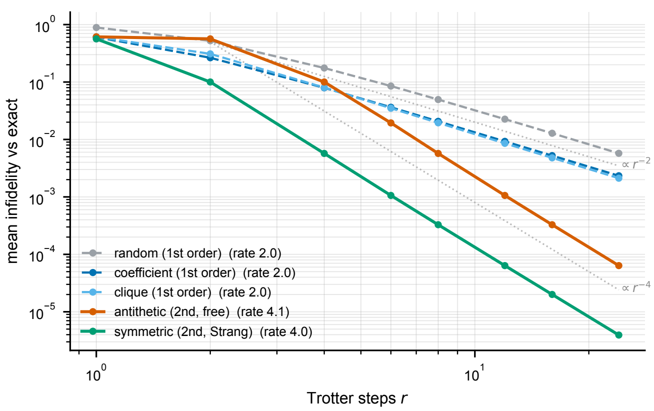

FIG. 3. **The antithetic schedule doubles the convergence order at the same gate count.** Mean infidelity against the exact propagator versus Trotter step count r (log–log), averaged over the 120 instances. The three first-order schedules (cool colors, dashed) lie nearly on top of one another at fitted rate ≈ 1.97 (parallel to the $\propto r^{-2}$ guide), the empirical signature of ordering impotence. The antithetic fold (vermillion), which uses the identical $L \cdot r$ rotations as the first-order schedules at each r, and the symmetric Strang fold (green) track the $\propto r^{-4}$ guide at rates 4.07 and 4.03.

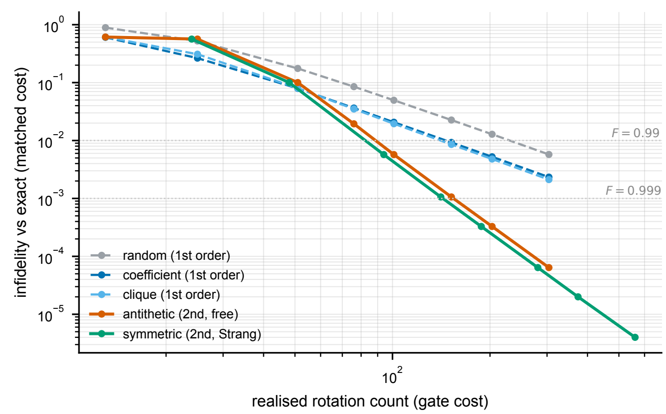

FIG. 4. **Matched-cost frontier: infidelity versus realized rotation count.** Mean infidelity (log) against the realized rotation count (log)—the matched-cost efficiency view—over the 120 instances. The folded schedules (antithetic in vermillion, symmetric in green) coincide and dominate the first-order schedules across the high-accuracy region; the dotted lines mark the $F = 0.99$ and $F = 0.999$ targets. First-order Trotter does not reach $F = 0.999$ within the swept budget; the folds do. The antithetic schedule achieves this at the *same* rotation count as the first-order schedules at each step count.

by roughly a third to a half and are the only schedules to reach 0.999. On *Heisenberg*, by contrast, the commuting-clique structure makes the first-order clique schedule essentially *exact* at a single step (Proposition 2): every clique-based schedule reaches 0.999 at the minimal rotation count, and the symmetric fold's 2× per-step overhead is here wasted. The gate-optimal schedule is therefore genuinely instance dependent—fold the hard instances, and leave the fast-forwardable ones first-order.

The learned scheduler of Sec. II F exploits this structure end-to-end. Trained leave-one-instance-out over the ten commutator-graph features, so that no test instance informs its own predictor, it attains 85% accuracy against the per-instance oracle. We note that, on the subset of 100 instances where every fixed schedule reaches the target, the scheduler needs 66.8 rotations on average, against an oracle optimum of 65.0 and the best single fixed schedules' 71.3 (symmetric) and 68.6 (antithetic)—recovering near-oracle efficiency by routing fast-forwardable instances to the cheap first-order schedule and hard instances to a fold. We emphasize that the

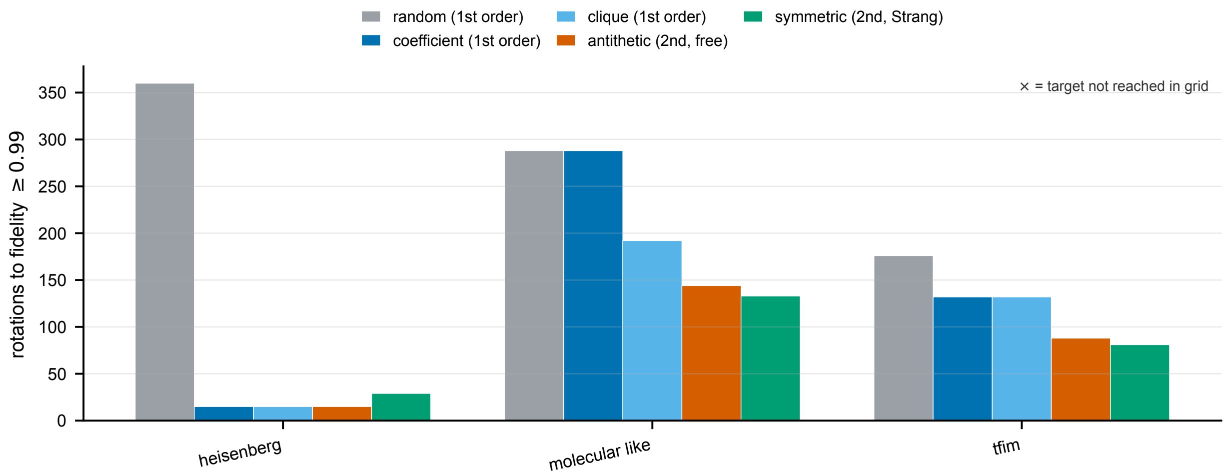

FIG. 5. **Rotations to fidelity** ≥ 0.99 **by Hamiltonian family.** Lower is better; an $\times$ marks a target not reached within the swept budget. On the dense molecular-like and transverse-field Ising families the folded schedules (antithetic, symmetric) reach the target at substantially fewer rotations than any first-order schedule. On Heisenberg the commuting-clique first-order schedules are already near-exact at the minimal budget, so folding only adds overhead. Bars are over the $N = 40$ instances per family.

TABLE IV. **Per-family rotations to a target fidelity.** For each family and schedule, the realized rotation count to reach mean fidelity ≥ 0.99 and ≥ 0.999; "—" marks a target not reached within the swept step grid. The folded schedules are the only ones to reach 0.999 on the molecular-like and transverse-field Ising families; the clique-based first-order schedules already reach it at minimal cost on Heisenberg. Transcribed from the `gates_to_target` block of `results/summary.json`.

family	schedule	$F \geq 0.99$	$F \geq 0.999$
heisenberg	random (1st)	360	—
	coefficient (1st)	15	15
	clique (1st)	15	15
	antithetic (2nd, free)	15	15
	symmetric (2nd)	29	29
molecular_like	random (1st)	288	—
	coefficient (1st)	288	—
	clique (1st)	192	—
	antithetic (2nd, free)	144	192
	symmetric (2nd)	133	177
tfim	random (1st)	176	—
	coefficient (1st)	132	—
	clique (1st)	132	—
	antithetic (2nd, free)	88	132
	symmetric (2nd)	81	121

scheduler is interpretable and listing-order invariant by construction (Proposition 3); it is a selector over named schedules, not a black-box policy.

IV. DISCUSSION

a. Summary of the result. In this manuscript we introduced commutator-graph scheduling (CGS), and the result it establishes has two halves that must be kept together. *Negatively*, a single term ordering is a provably weak lever for first-order Trotterization: reordering only re-signs the leading-error commutators, the leading-error norm is exactly invariant on collision-free families and within a few percent of its floor otherwise, and no ordering escapes the $O(t^2/r)$ rate—turning the well-known empirical observation that ordering heuristics are within noise into a theorem. *Positively*, the freedom that does matter is the per-step schedule: the antithetic fold cancels the leading error and converges at second order at the identical gate count, reaching a target fidelity with 1.62× fewer rotations and reaching 0.999 where first-order Trotter plateaus. We do not claim a new product-formula *primitive*—the symmetrization mechanism is classical [4, 7]—but rather the framing that the lever is the schedule, that a fixed ordering provably cannot realize it, and that the commutator graph both certifies the impotence and structures the schedule, together with the matched-cost quantification and the learned scheduler.

b. Relation to prior work. The commutator graph and its clique structure are established in measurement grouping for variational algorithms [12–14]; we repurpose them for *time evolution* and, beyond ordering, for scheduling. The sensitivity of Trotterized chemistry to term ordering has been reported empirically [10, 11, 15], and the impotence theorem explains the ceiling those studies encounter. Symmetric (Strang) and higher-order product formulas are classical [4, 7], and randomized compilation—random permutation per step and qDRIFT—improves error scaling by averaging rather than cancellation [8, 9]; the antithetic schedule is the *deterministic, zero-variance* counterpart that cancels the same leading term in a single sample, and our matched-cost frontier places it against both the first-order and the randomized baselines. The tight commutator-scaling theory of Trotter error [5, 6] is the lens throughout, and the commutator error form is its explicit, matrix-free, ordering-resolved representation.

c. Why a matched-cost schedule matters. For first-order formulas the schedule is genuinely free—it changes no gate, qubit, or depth count—so any accuracy it recovers is pure profit. In the high-accuracy regime where precise chemistry and materials simulation operate, the difference between an $O(t^2/r)$ and an $O(t^3/r^2)$ method at fixed cost is the difference between reaching and not reaching a target, as the unreachable 0.999 column of Table III shows. The commutator graph supplies both halves of the recommendation at no extra cost: it certifies when ordering cannot help, and its clique coloring tells the learned scheduler which instances to fold and which to leave alone.

d. Limitations. We report the boundaries of the study explicitly. (i) The second-order advantage is a small-step-size effect: at coarse steps the antithetic fold can be *worse* than a good first-order ordering (its curve in Fig. 4 starts above and crosses below near the intermediate budget); we report the crossover rather than hide it. (ii) The exact statevector reference requires forming $e^{-\mathrm{i}Ht}$, limiting us to $n \leq 8$ qubits, so the effect at scale—where a tensor-network or Krylov reference is needed—is not measured, although the convergence-rate argument is size independent. (iii) The families are transverse-field Ising, Heisenberg, and random local two-body molecular-like sums, a proxy for, not an instance of, mapped molecular electronic-structure Hamiltonians at scale [17, 18]. (iv) The cost is the realized rotation count and ignores the gate-synthesis cost of individual rotations and hardware connectivity; the antithetic schedule's seam rotations could be merged to reduce the count further, which we do not exploit so that its budget matches first-order exactly. (v) All results are classical statevector simulations on CPU, with no hardware validation.

e. Outlook. Most critically, because the antithetic upgrade changes no gate, qubit, or depth count, commutator-graph scheduling constitutes a zero-overhead accuracy improvement that any first-order Trotter compilation can adopt unchanged on near-term hardware. Natural next steps are larger systems with scalable references, real mapped molecular Hamiltonians, higher-order folded schedules, and a graph neural policy trained directly on a differentiable matched-cost objective. In short, by recasting the leading Trotter error as a certifiable operator on the commutator graph, CGS reframes term ordering as the weak lever it provably is, and identifies the per-step schedule as the free, structured degree of freedom that simulation should exploit.

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DATA AVAILABILITY

This study uses no external datasets. All Hamiltonians are generated synthetically from fixed random seeds by the accompanying code, and all numerical values, tables, and figures are produced by that code and stored in `results/summary.json`. The code that implements CGS and regenerates every figure, table, and reported value accompanies this manuscript as an installable package (`pip install .`) and will be released in a public repository upon publication.

Appendix A: Numerical methods and protocol

The analytic development—the Pauli algebra, the commutator error form, and the ordering-impotence, antithetic-cancellation, clique-exactness, and listing-order-invariance results—appears with complete proofs in Sec. II. Here we specify the numerical protocol. Three families are generated as $\{(c_j, P_j)\}$ term lists: transverse-field Ising $H = -J\sum_i Z_i Z_{i+1} - h\sum_i X_i$; Heisenberg $H = \sum_i (J_x X_i X_{i+1} + J_y Y_i Y_{i+1} + J_z Z_i Z_{i+1})$; and *molecular-like*, random local two-body Pauli terms with seeded Gaussian coefficients—a tractable stand-in for the dense, irregular anticommutation structure of Jordan–Wigner / Bravyi–Kitaev mapped electronic-structure Hamiltonians [17, 18]. The reported-scale configuration (`configs/full.yaml`) uses sizes $n \in \{4, 5, 6, 7, 8\}$, eight seeded instances per (family, size), evolution time $t = 1.0$, a reference budget of $r = 6$ steps, and a step grid $\{1, 2, 4, 6, 8, 12, 16, 24\}$ for the convergence and frontier curves, giving 120 instances in total. For each instance and schedule we evaluate the product formula, score fidelity and infidelity against the exact reference, and record the realized rotation count; from the (rotation count, fidelity) curves we read the rotations to each target fidelity and fit the convergence rate. Ordering impotence is measured by sampling 48 orderings per instance and forming the exact leading-error spectral norm. Per-schedule means carry 95% confidence intervals $1.96\,\hat{\sigma}/\sqrt{N}$.

Appendix B: Software, validation, and reproducibility

The pipeline is a CPU-only Python package (`topoham`) depending on NumPy [19], SciPy [20], NetworkX [21], scikit-learn [16], and Matplotlib. Hamiltonians are stored as Pauli term lists (`hamiltonians.py`); the Pauli algebra of Sec. II A—the $O(n)$ parity test (Lemma 1), the cosine–sine rotation kernel (Lemma 2), and the phased-Pauli product—is the inner loop of the exact statevector simulator (`pauli.py`). The commutator graph and matrix-free error form are built in `commutator_graph.py` and `error_form.py`: the greedy coloring partitions terms into commuting cliques (Proposition 2), the error form assembles Eq. (5) as a signed Pauli-string sum over the edges (Definition 1), and the same module computes the ten listing-order-invariant features the learned scheduler reads (Proposition 3). Schedules are realized as flat lists of (term, step-fraction) rotations whose seam-merged length is the gate cost on which all schedules are compared (`schedules.py`, `env.py`); the learned scheduler (`LearnedSchedulePolicy`) is the standardized logistic-regression classifier of Sec. II F, trained leave-one-instance-out.

The single source of truth is `results/summary.json`, written by the runner with full provenance (seed, command, platform, package versions, runtime, peak memory) and two integrity flags that must pass before any fidelity is produced: `pauli_algebra_verified` (the fast kernel matches dense `expm` to 10^{-9}) and `error_form_verified` (the matrix-free error form matches the dense pair-commutator sum to 10^{-9}). On the run reported here (seed 0, Python 3.13, macOS arm64), the experiment completed in $\approx 49\,\mathrm{s}$ wall-clock with $\approx 496\,\mathrm{MB}$ peak memory, both flags `true`. The macros, tables, and figures in this manuscript are generated directly from `summary.json`, so no number is entered by hand. The full pipeline is reproduced from the accompanying code by

```
pip install .
make test
export KMP_DUPLICATE_LIB_OK=TRUE
topoham-reproduce \
  --config configs/full.yaml
```

where `make test` checks the Pauli kernel (10^{-9}), the error form (10^{-9}), antithetic second-order convergence, and clique exactness, and `topoham-reproduce` regenerates `summary.json`, the tables, and the figures. All experiments are reproducible on commodity hardware.

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